Lab 6 – z-transform and LTI system in Transformed Domain

6.1 Introduction

The z-transform is useful for the manipulation of discrete data sequences and has acquired a new significance in the formulation and analysis of discrete-time systems. It is used extensively today in the areas of applied mathematics, digital signal processing, control theory, population science, and economics. These discrete models are solved with difference equations in a manner that is analogous to solving continuous models with differential equations. The role played by the z-transform in the solution of difference equations corresponds to that played by the Laplace transforms in the solution of differential equations.

6.2 z-transform

The z-transform of a time domain sequence, $x[n]$, is given by

$$X(z) = Z\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \tag{6.1}$$

where z is a complex variable. The set of z values for which $X(z)$ exists is called the region of convergence (ROC). If we express the complex variable z in polar form, $z = re^{j\omega}$, then (6.1) reduces to

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]r^{-n}e^{-j\omega n}, \tag{6.2}$$

which can be interpreted as the discrete-time Fourier transform of the modified sequence $x[n]r^{-n}$.

The inverse z-transform of a complex function $X(z)$ is given by:

$$x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}\,dz$$

where C is a counterclockwise contour encircling the origin and lying in the ROC.

6.2.1 Solving difference equation using z-transform

The transfer function of a generalized difference equation (or a generally infinite impulse response (IIR) system)

$$\sum_{m=0}^{N} b_m y[n-m] = \sum_{m=0}^{M} a_m x[n-m]$$

is a rational form $A(z)/B(z)$ given by

$$\frac{Y(z)}{X(z)} = \frac{\sum_{m=0}^{M} a_m\, z^{-m}}{\sum_{m=0}^{N} b_m z^{-m}} \triangleq H(z)$$

The ratio of the z-transform of the input signal to that of the output signal, $H(z)$, is defined to be the z-transform transfer function of the system.

Since LTI systems are often realized by difference equations, the rational form is the most common and useful form of z-transforms, represented as a rational function of z^{-1}

$$H(z) = \frac{P(z)}{D(z)} = \frac{p_0 + p_1 z^{-1} + \cdots + p_{M-1} z^{-(M-1)} + p_M z^{-M}}{d_0 + d_1 z^{-1} + \cdots + d_{N-1} z^{-(N-1)} + d_N z^{-N}} \tag{6.3}$$

where the degree of $P(z)$ is M, and that of $D(z)$ is N. The degree of the system is the larger one of M and N. When b_0 is normalized to 1, and $b_m = 0$ for $m = 1$ ... N, the difference equation degenerates to an FIR system.

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{m=0}^{M} a_m\, z^{-m}$$

It can still be represented by a rational form of the variable *z* as

$$H(z) = \frac{\sum_{m=0}^{M} a_m\, z^{(M-m)}}{z^M}$$

6.2.2 Poles and zeros of z-transform

The poles of $H(z)$ in rational form given in (6.3) are the roots of $D(z)$, and the zeros are the roots of $P(z)$, respectively. If we factorize $H(z)$ to the following form

$$H(z) = \frac{p_0}{d_0} \cdot \frac{\prod_{l=1}^{M} (1 - \xi_l z^{-1})}{\prod_{l=1}^{N} (1 - \lambda_l z^{-1})} = z^{(N-M)} \frac{p_0}{d_0} \cdot \frac{\prod_{l=1}^{M} (z - \xi_l)}{\prod_{l=1}^{N} (z - \lambda_l)},$$

the system contains *N* poles at $z = \lambda_l$, for $l = 1, 2, .., N$, and *M* zeros at $z = \xi_l$ for $l = 1, 2, .., M$.

Use the Matlab function **roots.m** to find the poles and zeros of the following z-transform

(a) $H_1(z) = -1 + 2z^{-1} - 3z^{-2} + 6z^{-3} - 3z^{-4} + 2z^{-5} - z^{-6}$

(b) $G_1(z) = \frac{3z^4 - 2.4z^3 + 15.36z^2 + 3.84z + 9}{5z^4 - 8.5z^3 + 17.6z^2 + 4.7z - 6}$

(c) $G_2(z) = \frac{2z^4 + 0.2z^3 + 6.4z^2 + 4.6z + 2.4}{5z^4 + z^3 + 6.6z^2 + 4.2z + 24}$

Use the Matlab function **zplane.m** to display the zero-pole plot. Determine all possible ROCs of each of the above z-transforms, and describe the type of their inverse z-transform (left-sided, right-sided, two sided sequences) associated with each of the ROCs.

INLAB REPORT: For each of the above z-transform, report the poles and zeros, hand in the zero-pole plots, show the all possible ROCs, and describe the type of their inverse z-transform associated with each of the ROCs.

6.2.3 *z*-transform and frequency response

In (6.2), if $r = 1(\text{i.e.}, |z| = 1)$, the *z*-transform reduces to DTFT. The contour $|z| = 1$ is a circle in the *z* plan of unity radius, called unit circle. If the ROC of $H(z)$ includes the unit circle, the frequency response $H(e^{j\omega})$ of the LTI digital system can be obtained by evaluating $H(z)$ on the unit circle:

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$$

For the digital filters characterized by the following transfer functions:

(a) $H(z) = z^{-4} + 2z^{-3} + 2z^{-1} + 1$, (b) $H(z) = \frac{z^2-1}{z^2-1.2z+0.95}$

Write a Matlab program to plot the magnitude (in both magnitude values and gain of dB) and phase of the frequency response of the digital filters with syntax [**mag**, **phase**]=**FreRes**(**num**, **den**), where **num**, and **den** are vectors specifying the coefficients of numerator and denominator polynomials in descending power of *z*, i.e., with **num**[1] and **den**[1] being the coefficient of z^0, **num**[2] and **den**[2] the coefficient of z^{-1}, etc.

INLAB REPORT: Submit your Matlab code and plots of the magnitude and phase responses of the transfer functions.

6.2.4 Inverse *z*-transform

In practice, inverse z-transforms are computed by avoiding contour integral if possible. One of the many methods is partial fraction expansion, in which, a fractional form z-transform

$$X(z) = \frac{P(z)}{D(z)} = \frac{\sum_{i=0}^{M} p_i z^{-i}}{\sum_{i=0}^{N} d_i z^{-i}}$$

is first written to a integer polynomial $K(z)$ plus a proper fraction, i.e.,

$$X(z) = K(z) + \frac{P_1(z)}{D(z)}$$

where the degree of $P_1(z)$ is less than *N*. Then, the proper fraction is expanded to partial fractions in the forms of

$$\frac{P_1(z)}{D(z)} = \sum_{l=1}^{N} \left(\frac{\rho_l}{1 - \lambda_l z^{-l}} \right)$$

where the $z = \lambda_k$, $1 \le k \le N$ are the poles of the proper fraction. In Matlab, the integer polynomial and the partial fractions together with the residues, ρ_l, maybe found by using function **residue.m**.

Read the help file of the function **residue.m**, and write a Matlab function to find the partial fraction expansion of the following two z-transforms:

(a) $X(z) = \frac{3-7.8z^{-1}}{(1-0.7z^{-1})(1+1.6z^{-1})}$, (b) $Y(z) = \frac{3z^2+1.8z+1.28}{(z-0.5)(z+0.4)}$

Each above z-transform has several ROCs. Evaluate their respective inverse z-transforms corresponding to each ROC.

INLAB REPORT: Submit the partial fraction expansion, all possible ROCs, and the corresponding inverse z-transforms.

6.2.5 Stability Conditions

A stable causal LTI system has all poles inside unit circle. This implies that a causal LTI FIR digital filter with bounded impulse response coefficients is always stable, as all its poles are at the origin in the z-plane. On the other hand, a causal LTI IIR digital filter may or may not be stable, dependent on its pole locations. In addition, an originally stable IIR filter characterized by infinite precision coefficients and with all poles inside the unit circle may become unstable after implementation due to the unavoidable quantization of all coefficients.

Slides 58 and 59 of L7 demonstrate an example where a stable IIR filter become unstable when the coefficients are quantized. Practice on the following:

1. Show the zero-pole plots of these two transforms, and observe the locations of the poles.
2. Matlab function **impz.m** can be used to compute a given number impulse response samples of digital filter with z-transform in rational form. Use **impz.m** to determine the first 100 samples of the impulse response of the two z-transforms.
3. Display and compare the two impulse response, and comment on their implications to the stabilities.

INLAB REPORT: Submit the zero-pole plots, and stem figure of the impulse responses, with your comments, if any.

6.3 Linear Phase FIR Filters

Linear Phase FIR filters are an important class of FIR filters. The frequency response of linear phase FIR filters is given by $e^{-j\left(\frac{N-1}{2}\omega-\beta\right)}R(\omega)$, where $R(\omega)$ is a real function, $\beta = 0\ or\ \pm 0.5\pi$ and N is the filter length, i.e., its phase function is in a linear form of $\theta(\omega) = \frac{N-1}{2}\omega - \beta$.

6.3.1 Four Types of Linear Phase FIR Filters

The linear phase of the FIR filter may be achieved by making the impulse response of the FIR filter either symmetric or anti-symmetric. Dependent on the parity of the filter length, there are 4 types of linear phase FIR filters.

Type I FIR filters: N is odd, and the impulse response is symmetric. Its frequency response is given by $H(e^{j\omega}) = e^{-j\omega\frac{N-1}{2}}\sum_{n=0}^{\frac{N-1}{2}} a[n]\cos(\omega n)$, where $a[0] = h\left[\frac{N-1}{2}\right]$, $a[n] = 2h\left[\frac{N-1}{2}-n\right]$, for $n = 1, 2, \ldots, \frac{N-3}{2}$. This type of filter has either an even number or no zeros at $z = 1$ and $z = -1$.

Type II FIR filters: N is even, and the impulse response is symmetric. Its frequency response is given by $H(e^{j\omega}) = e^{-j\omega\frac{N-1}{2}} \sum_{n=1}^{\frac{N}{2}} b[n]\cos(\omega(n - 1/2))$, where $b[n] = 2h[N/2 - n]$, for $n = 1, 2, \ldots, N/2$. This type of filter has either an even number or no zeros at $z = 1$ and an odd number of zeros at $z = -1$.

Type III FIR filters: N is odd, and the impulse response is anti-symmetric. Its frequency response is given by $H(e^{j\omega}) = je^{-j\omega\frac{N-1}{2}} \sum_{n=1}^{\frac{N-1}{2}} c[n]\sin(\omega n)$, where $c[n] = 2h\left[\frac{N-1}{2} - n\right]$, for $n = 1, 2, \ldots, \frac{N-1}{2}$. This type of filter has an odd number of zeros at $z = 1$ and $z = -1$.

Type IV FIR filters: N is even, and the impulse response is anti-symmetric. Its frequency response is given by $H(e^{j\omega}) = je^{-j\omega\frac{N-1}{2}} \sum_{n=1}^{\frac{N}{2}} d[n]\sin(\omega(n - 1/2))$where $d[n] = 2h[N/2 - n]$, for $n = 1, 2, \ldots, N/2$. This type of filter has an odd number of zeros at $z = 1$ and either an even number or no zeros at $z = -1$.

Besides the zeros at $z = 1$ and $z = -1$, all other zeros of real coefficient linear phase FIR filters exhibit mirror image symmetry with respect to unit circle.

An example of a type I linear phase FIR filter $H(z)$ has impulse response given by

$\{h[n]\}$ = {–0.0035, –0.0039, 0.0072, 0.0201, –0.0000, –0.0517, –0.0506, 0.0855, 0.2965, 0.4008, 0.2965, 0.0855, –0.0506, –0.0517, –0.0000, 0.0201, 0.0072, –0.0039, –0.0035}

for $0 \le n \le 18$.

Using the program developed in Section 6.2.3, plot the frequency response of the filter. What is the magnitude characteristic of the filter? Show as well the zero-pole plot of the filter.

INLAB REPORT: Submit the frequency response and zero-pole plots of the filter. Answer (1) the magnitude characteristic of the filter, and (2) if a type III filter may be designed to be a lowpass filter, and justify your answer.

The impulse response of the $h_1[n]$ and $h_2[n]$ is obtained by

$$h_1[n] = \begin{cases} h[n], & \text{for even } n \\ -h[n], & \text{for odd } n \end{cases}$$

$$h_2[n] = \begin{cases} h[n/5], & \text{for } n = 5k \text{ for interger } k \\ 0, & \text{otherwise} \end{cases}$$

INLAB REPORT: Express the z-transform of $h_1[n]$ and $h_2[n]$ in terms of $H(z)$, and express the frequency response of $H_1(e^{j\omega})$ and $H_2(e^{j\omega})$ in terms of $H(e^{j\omega})$. Verify the frequency response expressions by plotting the frequency response of $h_1[n]$ and $h_2[n]$. Describe the magnitude characteristic of the filter $H_1(z)$ and $H_2(z)$ with respect to that of $H(z)$.

6.3.2 Design of Simple FIR Filters

The simplest lowpass FIR filter is the first order moving average filter, with transfer function given by,

$$H_0(z) = \frac{1 + z^{-1}}{2}. \tag{6.4}$$

It is a type II filter with a zero at $z = -1$. The magnitude response of the filter has a maximum value one at $\omega = 0$, and a minimum value zero at $\omega = \pi$. The frequency response of the filter is given by,

$$H_0(e^{j\omega}) = e^{-j\omega/2}\cos(\omega/2).$$

The frequency $\omega = \omega_c$ at which $\left|H_0(e^{j\omega})\right| = \frac{1}{\sqrt{2}}\left|H(e^{j0})\right|$ is of practical interest as here the gain $\mathcal{G}(\omega_c)$ in dB is given by,

$$\mathcal{G}(\omega_c) = 20\log_{10}\left|H_0(e^{j\omega_c})\right| = 20\log_{10}\left|H_0(e^{j\omega_0})\right| - 20\log_{10}\sqrt{2} = -3.0103 \cong -3\text{dB}$$

Thus, the gain $\mathcal{G}(\omega_c)$ at $\omega = \omega_c$ is approximately 3-dB less than that at zero frequency. As a result, ω_c is called the 3-dB cutoff frequency. To determine the expression for ω_c, we set $\left|H_0(e^{j\omega_c})\right|^2 = 1/2$. For the filter in (6.4), it is $\cos^2(\omega_c/2) = 1/2$, yielding $\omega_c = \pi/2$.

A cascade of the simple filter of (6.4) results in an improved lowpass frequency response. The 3-dB cutoff frequency of a cascade of *M* sections of the lowpass filter of (6.4) is given by

$$\omega_c = 2\cos^{-1}\left(2^{-1/(2M)}\right). \tag{6.5}$$

Alternatively, improved lowpass frequency response can be obtained by a higher order moving average filter with transfer function given by

$$H(\mathrm{z}) = \frac{1}{M}\sum_{m=0}^{M-1} z^{-m}. \tag{6.6}$$

The frequency response of the moving average filter has been derived to

$$H(e^{j\omega}) = \frac{1}{M}\cdot\frac{\sin\left(\frac{M\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}e^{-\frac{j(M-1)\omega}{2}}.$$

Design a linear phase lowpass FIR filter with a 3-dB cutoff frequency at 0.3π using

(1) a cascade of first-order lowpass filters of (6.4). Determine the number of stages to be cascaded. Plot its gain response.

(2) a higher order moving average filter. Determine the order of the moving average filter.

Plot its gain response.

INLAB REPORT: Submit your derivation of (6.5) and the number of stages in the cascade design. Show the steps to determine the order of the filter in the higher order moving average filter design. Plot the gain responses of both designed filters on the same figure. Indicate the actual 3-dB cutoff frequency. Give your comments if any.

The simplest highpass FIR filter is obtained by replacing z with $-z$ in (6.4), resulting in a transfer function given by

$$H_1(z) = \frac{1 - z^{-1}}{2} \tag{6.7}$$

The corresponding frequency response is given by

$$H_1(e^{j\omega}) = je^{-j\omega/2}\sin(\omega/2)$$

Improved highpass frequency response can be obtained by cascading several sections of the simple highpass filter of (6.7), or replacing z with $-z$ in (6.6).

6.4 IIR Filters

The first-order lowpass IIR filter with a zero at $z = -1$ is given by

$$H_3(z) = \frac{1-\alpha}{2}\frac{1+z^{-1}}{1-\alpha z^{-1}},\ \ 0 < |\alpha| < 1$$

The squared magnitude function can be derived as,

$$\left|H_{LP}(e^{j\omega})\right|^2 = \frac{(1-\alpha)^2(1+\cos\omega)}{2(1+\alpha^2-2\alpha\cos\omega)}$$

and the 3-dB cutoff frequency may be determined as

$$\cos\omega_c = \frac{2\alpha}{1+\alpha^2}.$$

Therefore, when $-1 < \alpha < 0$, the resultant filter is a wide band lowpass filter (with the passband width larger than $\pi/2$), while $0 < \alpha < 1$, a narrow band lowpass filter (with the passband width narrower than $\pi/2$). For a given cutoff frequency ω_c, α is solved to be

$$\alpha = \frac{1-\sin\omega_c}{\cos\omega_c}$$

The first-order highpass IIR filter with a zero at $z = 1$ is given by

$$H_4(z) = \frac{1+\alpha}{2}\frac{1-z^{-1}}{1-\alpha z^{-1}},\ \ 0 < |\alpha| < 1 \tag{6.8}$$

Derive the squared magnitude response of the highpass IIR filter. Determine the 3-dB cutoff frequency, and for a given cutoff frequency ω_c, determine the value of α.

Design a first order highpass IIR filter with a 3-dB cutoff frequency of 0.2 π. Plot the magnitude response of the filter.

INLAB REPORT: Submit your derivation of the squared magnitude response, the 3-dB cutoff frequency, and α for a given cutoff frequency ω_c of the first order highpass IIR filter of (6.8). Design the highpass IIR filter, and plot the magnitude response of the resultant filter. Indicate the actual 3-dB cutoff frequency. Give your comments if any.